

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1

Year: 2020

Subject: 341151 Electric Circuits I

Section: 5-6

Date: 8 September 2020

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 11 pages (including this page) and they are worth 65 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test papers.
- Books, documents, and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (15 points) True/False

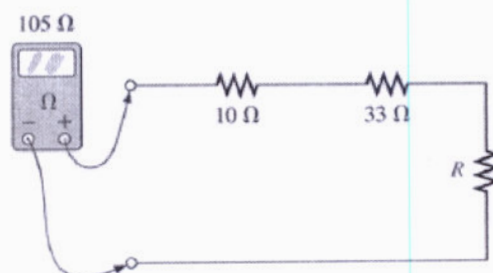
Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. The notation μs is an abbreviation for milli-seconds.
2. The prefix that corresponds to 10^9 is Tera.
3. From metric standard, CGS represents Centimeters, Grams, and Seconds.
4. Volt is equivalent to Joule per Coulomb.
5. The SI unit of inductance is named after Michael Faraday.
6. Conventional current is in the same as the direction of electron current.
7. The unit of conductance is in Ohm.
8. From Coulomb's law $F = k \frac{Q_1 Q_2}{r^2}$, where k denotes a constant of 9×10^9 .
9. Kirchhoff's current law requires that total current input to a system must equal to the total current output from the system.
10. Kirchhoff's voltage law requires that total voltage input to a system must equal to the total voltage output from the system.
11. One square mil equals to $4/\pi \text{ cm}^2$.
12. $\eta = \frac{P_o}{P_{in}}$ denotes the power efficiency of a system.
13. The circuit breaker is designed to protect short circuit.
14. Internal resistance of the voltmeter is in Mega ohm range, thus it is not affected to the circuit.
15. The heat from a microwave oven occurs due to the friction of the water molecules in the food.

Part 2 (30 points) Multiple choices

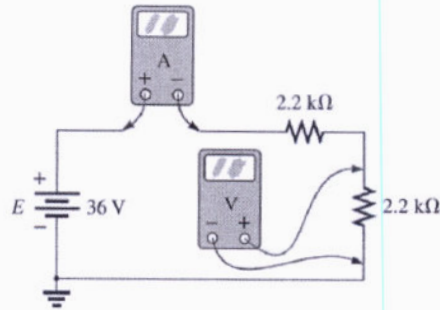
Instruction: Mark the correct answer for following questions in the provided answer sheet.

- The current divider rule states that for two parallel _____ of equal value, the current will be divided equally.
 - elements
 - resistors
 - circuits
 - branches
- Four 3-volt batteries connected in series will result in which of the following voltage?
 - 3 volts
 - 6 volts
 - 12 volts
 - 48 volts
- In double-subscript notation, the voltage V_{ba} is the same as
 - $V_a - V_b$
 - $V_a + V_b$
 - $V_b - V_a$
 - V_{ab}
- Evaluate the following expression: $(1 \times 10^{-6}) / (1 \times 10^{-9}) =$
 - 1×10^{-3}
 - 1×10^3
 - 0.01
 - 100
- From Figure below, the value of R is



- 43 Ω
- 62 Ω
- 82 Ω
- 148 Ω

6. From Figure below, which of the following gives the value of voltmeter reading?



- (a) 12V (b) 18V (c) 36V (d) 48V

7. From previous problem, which of the following gives the value of ammeter reading?

- (a) 0 mA (b) 2.2 mA (c) 8.18 mA (d) 16.36 mA

8. Potential is defined as the _____ at a point with respect to another point in the electric circuit.

- (a) charge (b) current
(c) ampere (d) voltage

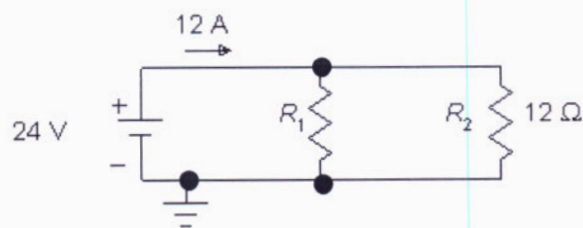
9. For how many hours will a 9 V battery with an ampere rating of 320 mAh theoretically provide a current of 15.7 mA?

- (a) 18.3 h (b) 17.3 h (c) 19.3 h (d) 20.4 h

10. What is the cost of using a 24 W radio for 1 hour at a rate of 2 Bahts per kilowatthour?

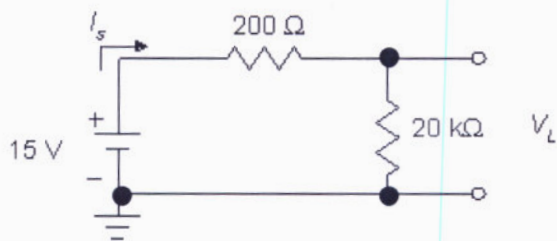
- (a) 0.024 Baht (b) 0.24 Baht (c) 0.048 Baht (d) 0.48 Baht

11. Determine the resistance R_1 .



- (a) $0.24\text{ }\Omega$ (b) $240\text{ }\Omega$ (c) $24\text{ }\Omega$ (d) $2.4\text{ }\Omega$

12. Determine I_s .



- (a) 743 mA (b) 0.743 A (c) 0.743 mA (d) 7.43 mA

13. From previous problem, determine V_L if the $20\text{ k}\Omega$ resistor is opened circuit.

- (a) 7.5 V (b) 0 V (c) -7.5 V (d) 15 V

14. Find the energy to move charge of 10 C between two points if the voltage is 50 V.

- (a) 500 J (b) 50 J
(c) 5 J (d) 0.5 J

15. What is the efficiency of a motor that has an output of 1.5 hp with an input of 1550 W?

- (a) 0.722 (b) 7.22 (c) 0.139 (d) 1.39

16. Which of the following expresses a double-superscript notation of $V_{ab} = 15\text{ V}$?

- (a) Point a has a higher potential than b.
(b) Point a has the same potential as b.
(c) Point a has a lower potential than b.
(d) Point a cannot be determined

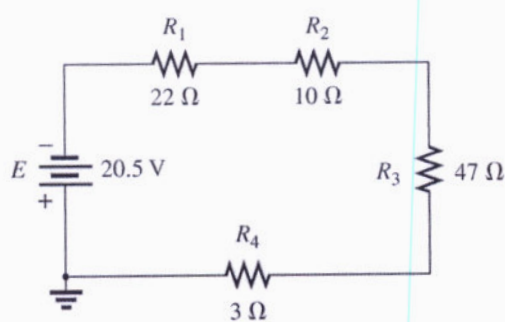


Figure 1.

17. See Figure 1. The total resistance in this circuit is

- (a) $82\ \Omega$ (b) $22\ \Omega$
(c) $100\ \Omega$ (d) $10\ \Omega$

18. See Figure 1. The total current flowing from the battery is

- (a) 2.5 mA (b) 0.25 mA
(c) 2.5 A (d) 0.25 A

19. See Figure 1. The total power dissipated by the $47\ \Omega$ resistor is

- (a) 11.75 W (b) 2.93 W
(c) 1.33 mW (d) 8.94 W

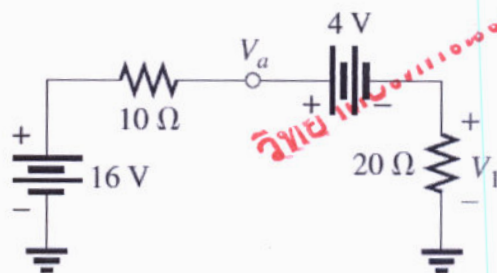


Figure 2.

20. See Figure 2. The total resistance in this circuit is

- (a) $10\ \Omega$ (b) $20\ \Omega$
(c) $30\ \Omega$ (d) $40\ \Omega$

21. See Figure 2. The total current is

- (a) +500 mA (b) -500 mA
(c) +2.5 A (d) -2.5 A

22. A circuit has a source voltage of 15 volts and two resistors with a total resistance of 4000 ohms. If R_1 has a potential of 9.375 volts, what is the value of R_2 ?

- (a) 500 ohms (b) 1000 ohms
(c) 1500 ohms (d) 2000 ohms

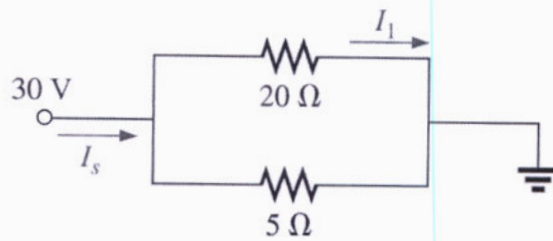


Figure 3.

23. See Figure 3. What is the current I_s ?

- (a) -7.5 A (b) +7.5 A
(c) -1.2 A (d) +1.2 A

24. See Figure 3. What is the current I_1 ?

- (a) -6.0 A (b) +6.0 A
(c) -1.5 A (d) +1.5 A

25. The attraction force between two charges which can be determined by _____.

- (a) Coulomb's law (b) Newton's law
(c) Kirchhoff's current law (d) Ohm's Law

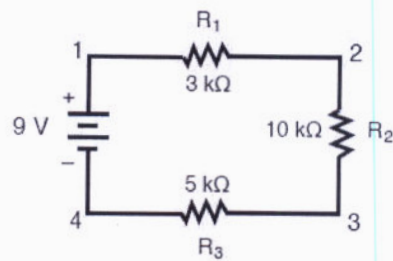
26. How much charges pass through a battery of 4.5 V if the energy expended is 60 J?

- (a) 13.5 C (b) 13.3 C (c) 10.6 C (d) 11.4 C

27. What is the total cost of using 1,000 W air conditioner for 12 h at a rate of 10 Baht per kilowatthour?

- (a) 160 Bahts (b) 120 Bahts (c) 12.0 Bahts (d) 240 Bahts

28. Following the circuit below, find the current I .

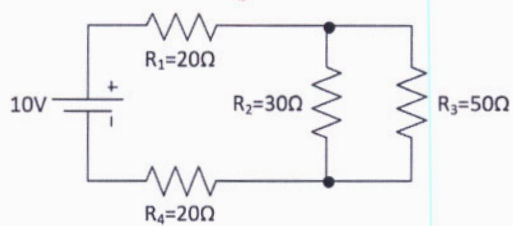


- (a) 1.5 mA (b) 0.5 mA (c) 1.8 mA (d) 18.5 mA

29. Find the dissipated power of R colored green, red, blue, gold and gold, where the current is 1 A.

- (a) 526 W (b) 52.6 W (c) 5.26 W (d) 0.526 W

30. Calculate R_T

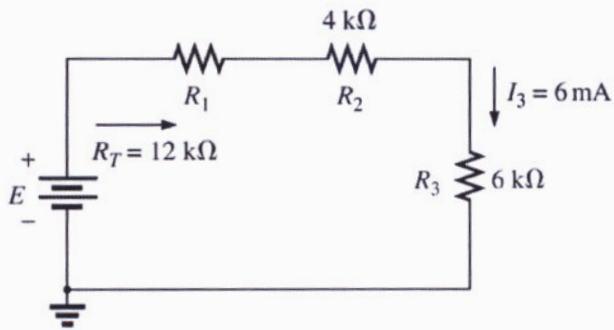


- (a) 38.75 Ω (b) 58.75 Ω (c) 70 Ω (d) 90 Ω

Part 3 (20 points) Calculation

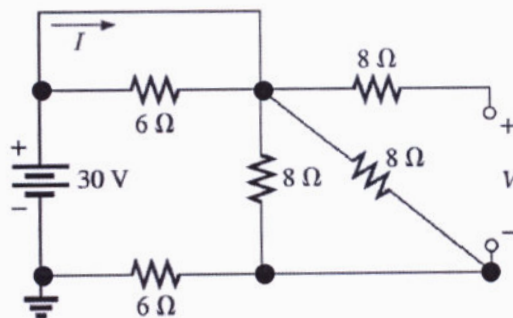
Instruction: Show the mathematical expression and answers of following problems.

1. (5 points) Find the value of E and R_1 .

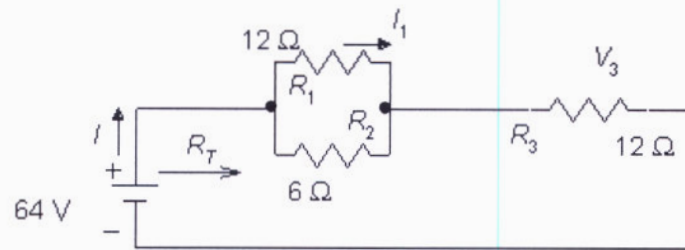


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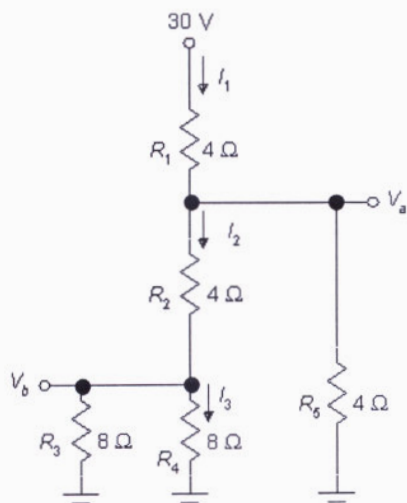
2. (5 points) From Figure below, find the voltage V and current I .



3. (5 points) Calculate I_1 , V_3 and R_T .



4. (5 points) From Figure below, determine the current I_1 , I_2 , I_3 , and voltage V_a , V_b



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Asst. Prof. Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 341151 Electric Circuits I

Part 1

	True	False
1.		
2.		
3.		
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Part 2

	a	b	c	d
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